ZHAN YU

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Ph.D. Candidate in Quantum Computing
Centre for Quantum Technologies, NUS, Singapore

RESEARCH INTERESTS

My research interests lie in quantum computing and quantum information, with two main focuses. The first is to design efficient quantum algorithms that provide advantages over classical methods. The second is quantum learning theory, with the goal of learning and characterizing quantum systems. I am also broadly interested in theoretical computer science.

EDUCATION

National University of Singapore, Singapore — Sep. 2023 – Present
Ph.D. in Quantum Computing
Supervisor: Dr. Patrick Rebentrost

University of Calgary, Calgary, AB — Sep. 2019 – Sep. 2021
M.Sc. in Computer Science
Thesis: Quantum walks on arc-transitive graphs with self-loops
Supervisor: Dr. Peter Høyer
GPA: 3.93/4.00

University of Calgary, Calgary, AB — Sep. 2016 – Jun. 2019
B.Sc. Honours in Computer Science (Minor in Pure Mathematics)
Thesis: Efficient quantum walk on the grid using self-loops
Supervisor: Dr. Peter Høyer
GPA: 3.97/4.00

Wuhan University of Technology, Wuhan, Hubei — Sep. 2014 – Jun. 2016
B.Eng. in Software Engineering
GPA: 88.12/100

RESEARCH EXPERIENCE

School of Mathematics and Statistics, *Wuhan University* — May 2023 – Aug. 2023
Visiting Student
Host: Dr. Yinan Li
Conducted research in quantum algorithms and quantum learning theory, focusing on the expressive power and generalization of quantum learning models.

Institute for Quantum Computing, *Baidu Research* — Nov. 2021 – May 2023
Research & Development Intern
Mentor: Dr. Xin Wang

- Conducted research in quantum algorithms, quantum information, and quantum learning theory, resulting in peer-reviewed publications and more than 20 patents. Characterized the expressive power of quantum neural networks and extended the framework to multi-qubit quantum phase processing, yielding intuitive and efficient algorithms for phase-related problems.
- Contributed to Paddle Quantum and Quantum Leaf, platforms for designing, testing, and deploying programs on quantum simulators and cloud-accessible quantum hardware. Implemented modules and tutorials for variational ansatzes, quantum singular value transformation, and quantum phase processing.

Institute for Quantum Science and Technology, *University of Calgary* Apr. 2018 – Sep. 2021
Research Assistant
Supervisor: Dr. Peter Høyer

- Analyzed the time complexity and success probability of lackadaisical quantum-walk search, proving several prior numerical observations and an open conjecture.
- Contributed to a MATLAB toolbox for efficiently simulating quantum walks on multiple graph families and calculating quantum hitting times and success probabilities.

TEACHING EXPERIENCE

Department of Computer Science, *University of Calgary* Sep. 2019 – May 2021
Teaching Assistant
DATA 211 - Programming with Data
CPSC 217 - Introduction to Computer Science for Multidisciplinary Studies I
CPSC 231 - Introduction to Computer Science for Computer Science Majors I
CPSC 519/619 - Introduction to Quantum Computation

PUBLICATIONS

(*) after my name denotes co-first authorship or alphabetical author order. An up-to-date publication list is available on Google Scholar.

1. Mingrui Jing, Xin Wang, **Zhan Yu***, and Lei Zhang, *Multivariate Quantum Signal Processing with Optimal Query Complexity*, Oct. 2026, arXiv: 2610.03393 [quant-ph]
2. Paolo Recchia, **Zhan Yu**, Kelvin Koor, and Patrick Rebentrost, *Quantum Quasi-Monte Carlo: A Window for Pre-Asymptotic Quantum Advantage*, Sept. 2026, arXiv: 2609.03625 [quant-ph]
3. Yupan Liu, Qisheng Wang, and **Zhan Yu***, "On Estimating Operator Norm Distance, with Optimal Trace Distance Estimation When One State Is Pure", in: *34th Annual European Symposium on Algorithms (ESA 2026)*, vol. 388, Leibniz International Proceedings in Informatics (LIPIcs), Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Aug. 2026, 71:1–71:18, arXiv: 2607.03905 [quant-ph]
4. Zhili Chen and **Zhan Yu***, *Learning Arbitrary Lindbladians from Time Evolution*, July 2026, arXiv: 2607.28610 [quant-ph]
5. Itai Arad, Zhili Chen, Naixu Guo, Patrick Rebentrost, and **Zhan Yu***, *Near-Optimal Learning of Local Lindbladians*, June 2026, arXiv: 2606.20535 [quant-ph]
6. Kean Chen, Qisheng Wang, **Zhan Yu***, and Zhicheng Zhang, "Simultaneous Estimation of Nonlinear Functionals of a Quantum State", *IEEE Transactions on Information Theory* 72.9, Sept. 2026, pp. 6740–6751, arXiv: 2505.16715 [quant-ph]
7. Yuxuan Du, Xinbiao Wang, Naixu Guo, **Zhan Yu**, Yang Qian, Kaining Zhang, Min-Hsiu Hsieh, Patrick Rebentrost, and Dacheng Tao, *A Gentle Introduction to Quantum Machine Learning*, Singapore, Springer Nature Singapore, 2025
8. J.-T. Bu, Lei Zhang, **Zhan Yu**, Jing-Bo Wang, W.-Q. Ding, W.-F. Yuan, B. Wang, H.-J. Du, W.-J. Chen, L. Chen, J.-W. Zhang, J.-C. Li, F. Zhou, Xin Wang, and M. Feng, "Exploring the Experimental Limit of Deep Quantum Signal Processing Using a Trapped-Ion Simulator", *Physical Review Applied* 23.3, Mar. 2025, p. 034073
9. Chengkai Zhu, Yu-Ao Chen, Zanqiu Shen, Zhiping Liu, **Zhan Yu**, and Xin Wang, *Amortized Stabilizer Rényi Entropy of Quantum Dynamics*, Sept. 2024, arXiv: 2409.06659 [quant-ph]
10. Naixu Guo, **Zhan Yu**, Matthew Choi, Yizhan Han, Aman Agrawal, Kouhei Nakaji, Alán Aspuru-Guzik, and Patrick Rebentrost, *Quantum Transformer: Accelerating Model Inference via Quantum Linear Algebra*, Feb. 2024, arXiv: 2402.16714 [quant-ph]
11. **Zhan Yu**, Qiuhao Chen, Yuling Jiao, Yinan Li, Xiliang Lu, Xin Wang, and Jerry Zhijian Yang, "Non-Asymptotic Approximation Error Bounds of Parameterized Quantum Circuits", in: *Advances in Neural Information Processing Systems*, vol. 37, Curran Associates, Inc., 2024, pp. 99089–99127

12. Youle Wang, Lei Zhang, **Zhan Yu**, and Xin Wang, "Quantum Phase Processing and Its Applications in Estimating Phase and Entropies", *Physical Review A* 108.6, Dec. 2023, p. 062413
13. Yifei Chen, **Zhan Yu**, Chenghong Zhu, and Xin Wang, *Efficient Information Recovery from Pauli Noise via Classical Shadow*, May 2023, arXiv: `2305.04148 [math-ph, physics:quant-ph]`
14. **Zhan Yu**, Xuanqiang Zhao, Benchi Zhao, and Xin Wang, "Optimal Quantum Dataset for Learning a Unitary Transformation", *Physical Review Applied* 19.3, Mar. 6, 2023, p. 034017
15. **Zhan Yu**, Hongshun Yao, Mujin Li, and Xin Wang, "Power and Limitations of Single-Qubit Native Quantum Neural Networks", in: *Advances in Neural Information Processing Systems*, vol. 35, Curran Associates, Inc., 2022, pp. 27810–27823
16. Peter Høyer and **Zhan Yu***, "Analysis of Lackadaisical Quantum Walks", *Quantum Information and Computation* 20.13–14, Nov. 2020, pp. 1137–1152

SELECTED TALKS

1. *Simultaneous Estimation of Nonlinear Functionals of a Quantum State*. 2025
Contributed talk at the 25th Asian Quantum Information Science Conference (AQIS 2025, Hong Kong).

2. *Non-Asymptotic Approximation Error Bounds of Parameterized Quantum Circuits*. 2024
Spotlight presentation at the 38th Conference on Neural Information Processing Systems (NeurIPS 2024, online); short oral presentation at the 8th Conference on Quantum Techniques in Machine Learning (QTML 2024, Melbourne).

3. *Power and limitations of single-qubit native quantum neural networks*. 2022
Presented at the 36th Conference on Neural Information Processing Systems (NeurIPS 2022, online).

AWARDS & SCHOLARSHIPS

Outstanding Intern Award 2022, *Baidu Research* Dec. 2022

Alberta Innovates Graduate Student Scholarships 2020, *University of Calgary* Nov. 2020

Alberta Graduate Excellence Scholarship (AGES) 2020, *University of Calgary* Jul. 2020

Alberta Graduate Excellence Scholarship (AGES) 2019, *University of Calgary* Nov. 2019

International Graduate Recruitment Award 2019, *University of Calgary* Sep. 2019

Visa Differential Scholarship, *University of Calgary* Sep. 2019

International Undergraduate Award 2018, *University of Calgary* Dec. 2018

SUPER Work Award 2018, *University of Calgary* Sep. 2018

George Loo Scholarship in Computer Science, *University of Calgary* Jul. 2018

Program for Undergraduate Research Experience (PURE) Award, *University of Calgary* May 2018

Transfer Student Scholarship, *University of Calgary* Sep. 2016

Third-Tier Scholarship, *Wuhan University of Technology* Oct. 2015

ACADEMIC SERVICE

Journal reviewer: *Quantum*, *Physical Review Research*, and *Physical Review A*

Conference reviewer: TQC, AQIS, and QIP